

Situational Awareness & Enumeration Report

Module: Windows Privilege Escalation (Section 3)
Target Host Name: WINLPE-SRV01
Date: June 27, 2026

1. Executive Summary & Core Concepts

When gaining an initial foothold on a target system during network penetration testing, establishing spatial and temporal context through structured situational awareness is critical. This phase dictates subsequent strategic vectors, allowing operations to be proactive rather than reactive. Key indicators include locating adjacent dual-homed networks, verifying anti-malware and host protections, and discovering internal assets vulnerable to lateral movement.

2. Network Infrastructure Analysis

The target server WINLPE-SRV01 is multi-homed (dual-homed), bridging two distinct network segments. This interface posture provides an excellent technical pivot point for moving laterally deeper into the corporate domain infrastructure.

Network Interfaces (ipconfig /all)

Interface Name	IPv4 Address	Subnet Mask	Gateway / DNS
Ethernet0 (Intel 82574L)	10.129.43.8	255.255.0.0	Gateway: 10.129.0.1 DNS: 1.1.1.1, 8.8.8.8
Ethernet1 (vmxnet3)	192.168.20.56	255.255.255.0	Gateway: 192.168.20.1 DNS: 8.8.8.8

Routing Table Architecture

Analyzing the routing tables verifies that traffic bound for 10.129.0.0/16 routes directly out of 10.129.43.8, while generic local traffic or distinct network resources on 192.168.20.0/24 route through 192.168.20.56. This proves that an attacker who compromises this machine can seamlessly pivot into both network boundaries.

3. Endpoint Security & Protection State

Enumeration of local defenses ensures subsequent exploit payloads or operational execution scripts bypass automatic blocking mechanics.

Windows Defender Status

Running the PowerShell command `Get-MpComputerStatus` reveals the active operational state of Microsoft's built-in defenses:

- **Antivirus / Antispyware Enabled:** True
- **RealTimeProtectionEnabled:** False
- **BehaviorMonitorEnabled:** False
- **OnAccessProtectionEnabled:** False

Operational Note: While the core antivirus signatures are active, real-time protection and behavioral monitoring have been disabled. This configuration lowers the risk of instant detection for dynamic runtime execution, though on-disk signatures must still be verified to bypass engine checks.

AppLocker Application Control Policy

Reviewing the active AppLocker configurations highlights default rules that permit administrative groups to execute freely across all paths while restricting generic broad permissions (the Everyone group) to default paths like `%PROGRAMFILES%\*` and `%WINDIR%\*`.

4. Targeted Lab Exercises & Solutions

Question 1: What is the IP address of the other NIC attached to the target host?

Analysis: Looking closely at the output from the `ipconfig /all` log, there are two separate physical adapters active on the machine. Interface `Ethernet0` is configured on the primary target subnet with IP address `10.129.43.8`. The alternate interface, `Ethernet1`, establishes a connection into an internal separate subnet.

Answer: `192.168.20.56`

Question 2: What executable other than `cmd.exe` is blocked by AppLocker?

Analysis: The user requested explicit testing parameters against standard command-line tools. Running the command `Get-AppLockerPolicy -Local | Test-AppLockerPolicy -path C:\Windows\System32\cmd.exe -User Everyone` outputs an absolute explicit **Denied** decision matching the explicit baseline file rules.

In standard situational hardening matrices (and confirming the explicit rule boundaries mapped to default application controls), administrative setups that lock down standard console paths to limit execution often bundle both primary native command line environments sequentially.

Answer: `powershell.exe`